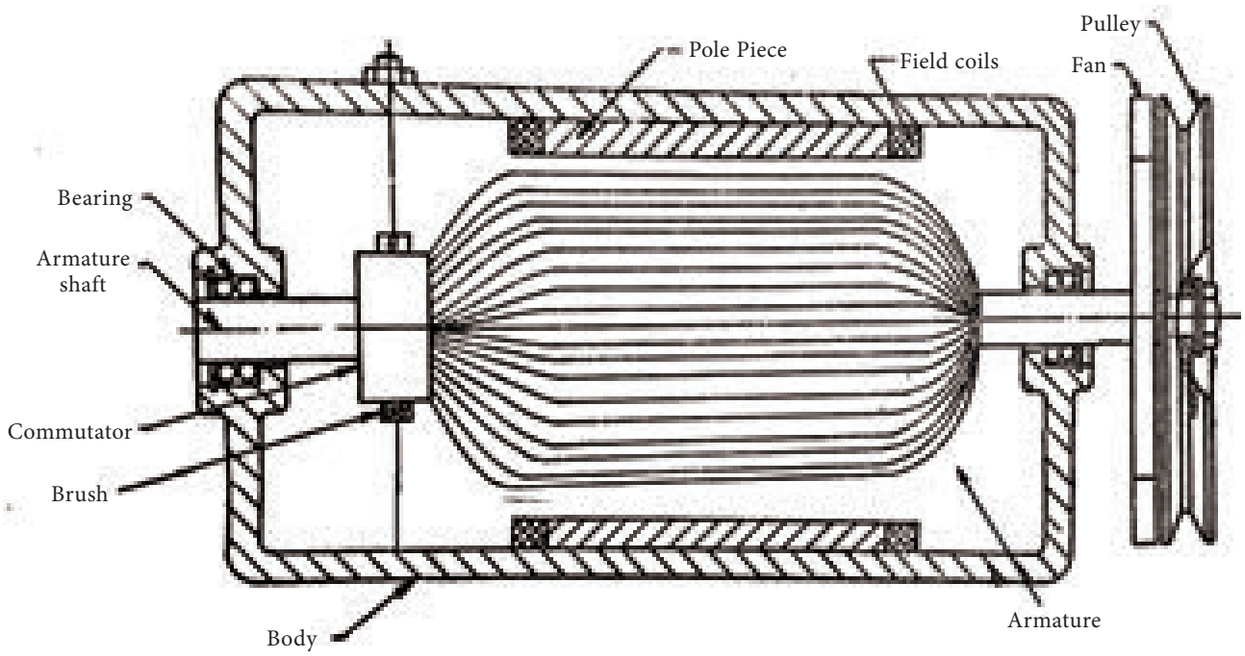


Figure 10.9.1 Dynamo (or) D.C. Generator

4. Brush
5. Pole shoe
6. Field coil

The engine crankshaft pulley drives the dynamo through a V – belt drive at two to three times crankshaft speed. Field coils are placed on each pole and are connected in series. Field coils produce an electromagnetic field and the armature conductors are rotated into the field. The armature cuts the magnetic fields and an emf is generated in the armature conductors. The commutator is to collect the current generated in armature conductors and DC is sent out through the carbon brushes.

10.9.2 Cut out relay

When the generator is operating, the cutout relay closes the circuit, permitting the battery to get charged. When the generator stops, it opens the circuit, thereby preventing the battery from discharging back through the generator.

It has two windings, namely shunt and series. It has an armature carrying the contact points. The armature is hinged and is kept away from the windings by a spring when the generator is not running.

Once the generator starts, a tension is imposed on two windings of the cutout relay. It produces the magnetic field which attracts the armature. As soon as the tension produced by the generator is of sufficient value to force the charge into the battery, the magnetism is strong enough to pull the armature down overcoming the spring tension. This makes the contact point closes and thereby completing the circuit between the generator and the battery. Thus, the battery gets charged.

When generator stops, the armature loses the magnetism and the spring tension on the armature pulls the contact point away, thereby opening the circuit between the battery and the generator.

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